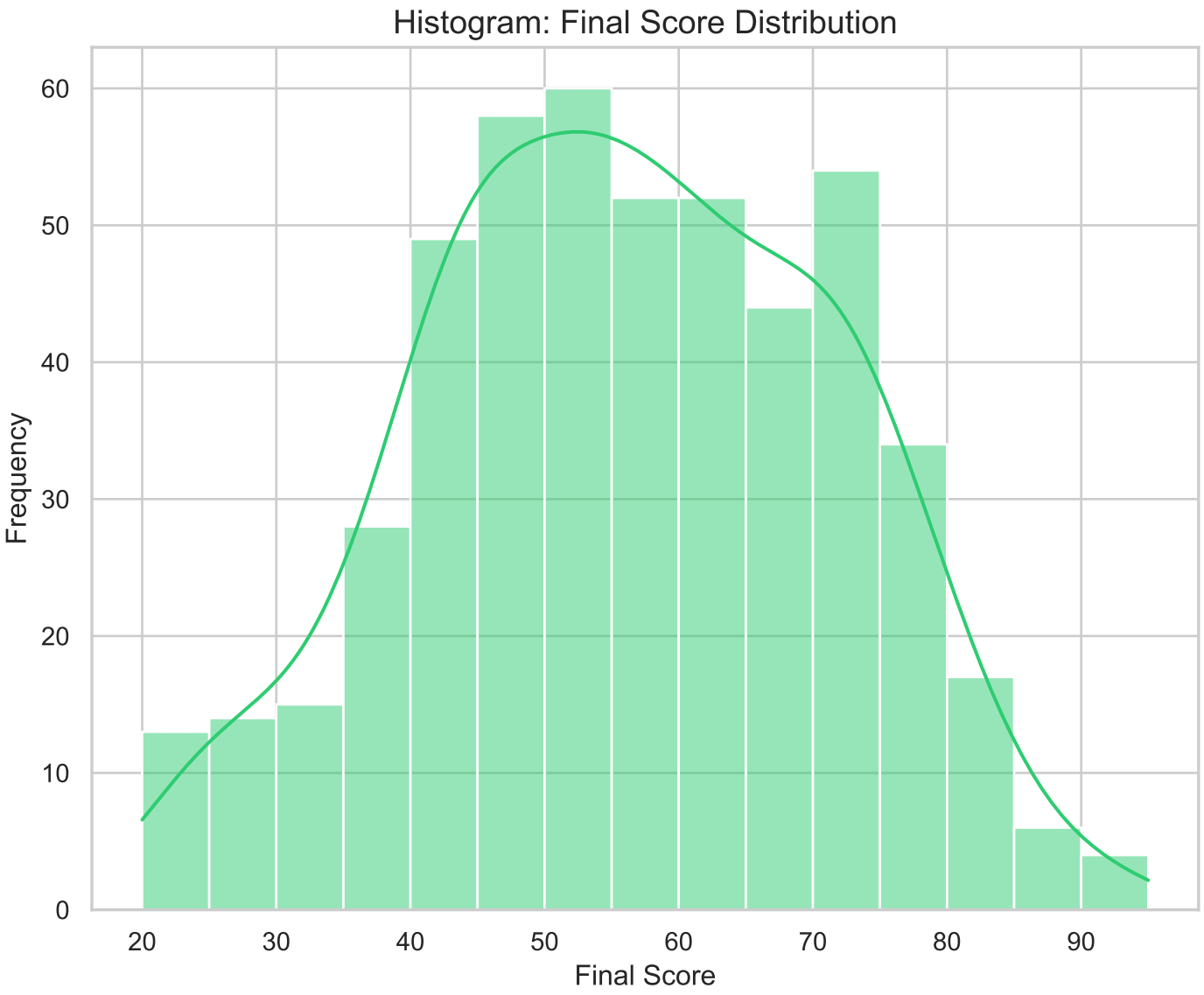
College Analytics Report
Mid-Term Grades Analysis

Automated Statistical Insights and Data Visualizations

Generated by Statistical Explorer v1.0
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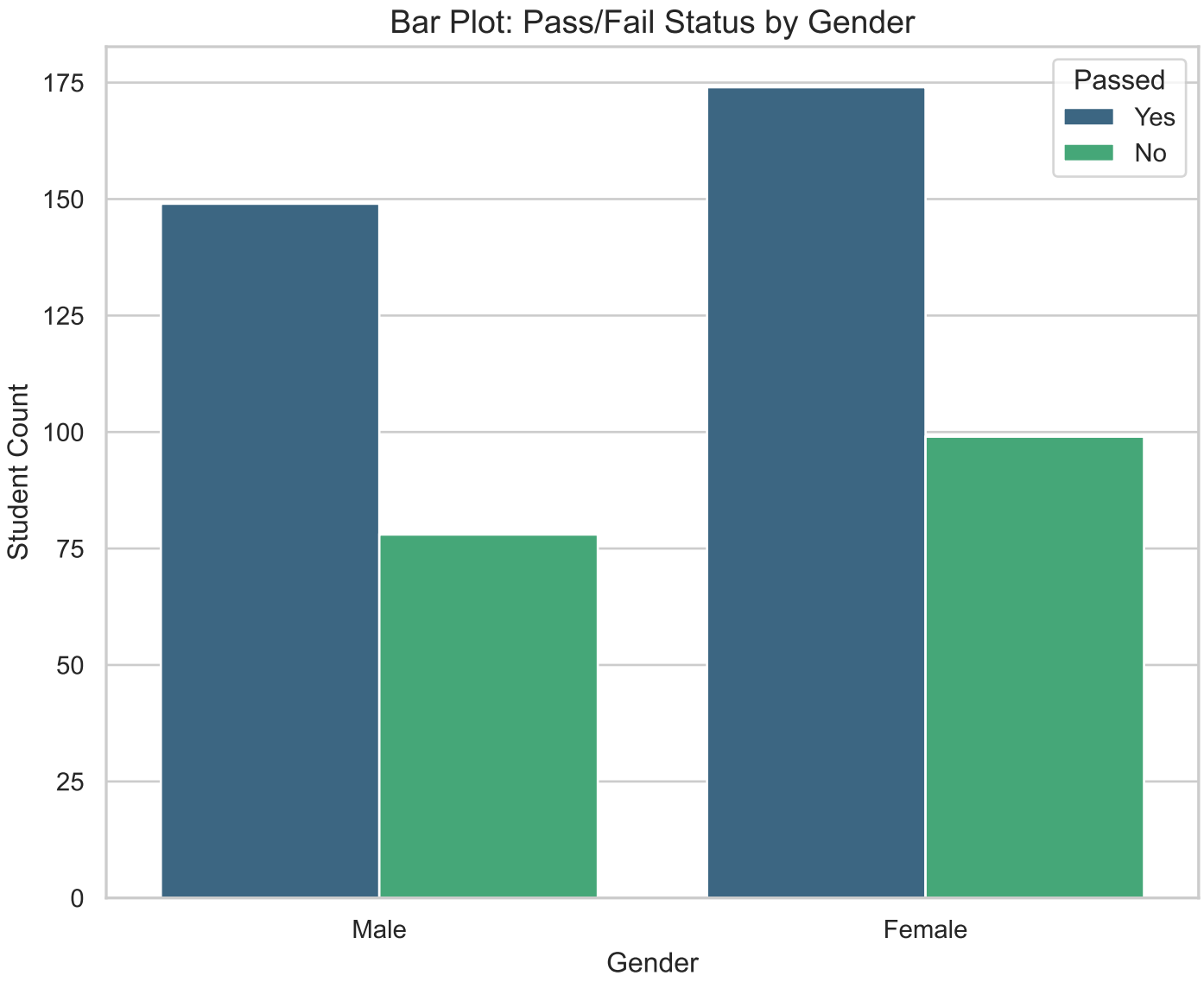
1. Distribution of Final Scores

This histogram illustrates the frequency distribution of final scores among students. It helps identify the overall performance spread, indicating whether grades are distributed normally or skewed toward specific ranges. The overlaid KDE (Kernel Density Estimate) line provides a smooth representation of the data density, highlighting the most common score ranges.



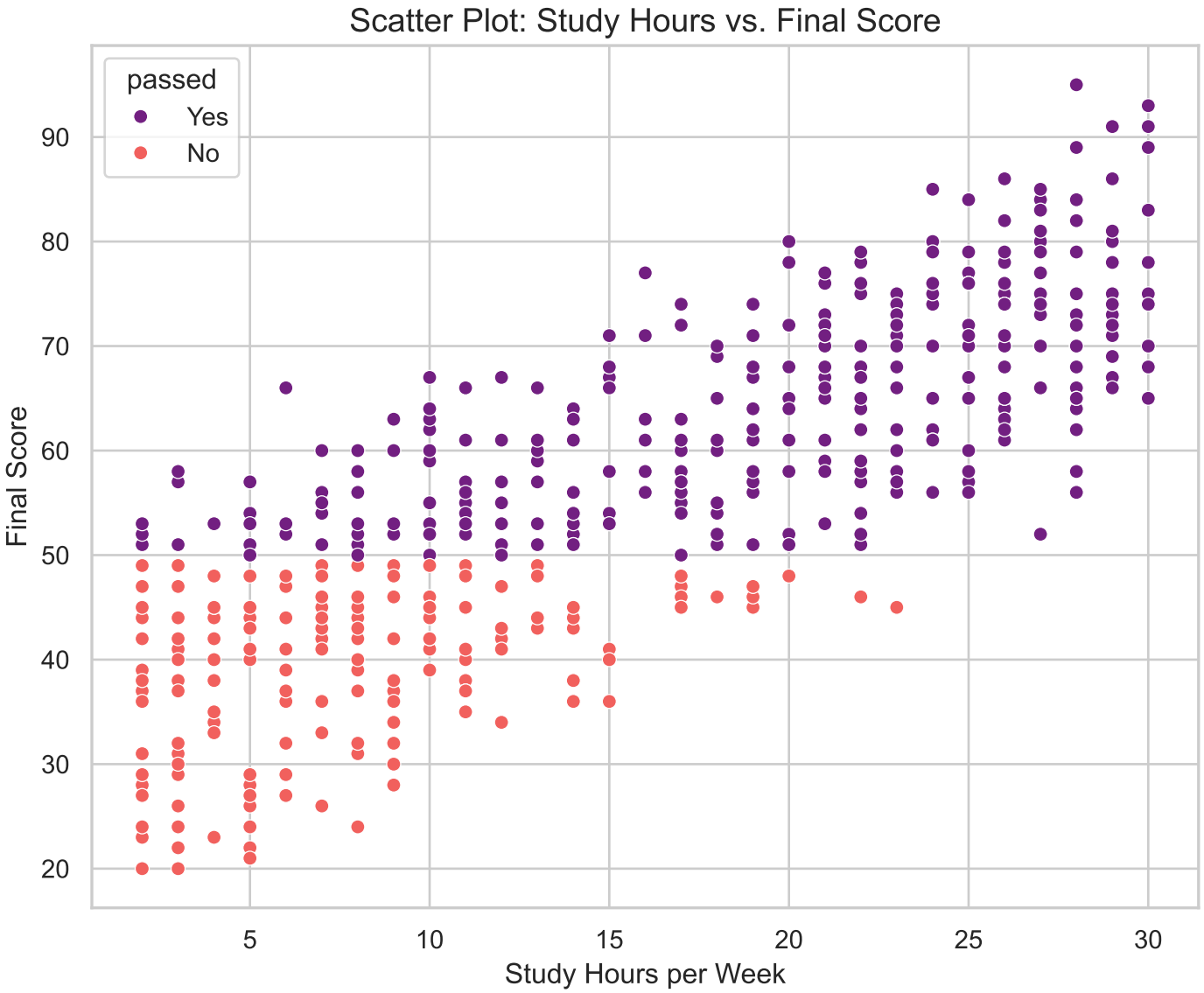
2. Performance by Gender

This bar chart compares the count of students who passed or failed, categorized by gender. This visualization is crucial for identifying any gender-based performance trends or gaps. By observing the ratio of 'Yes' to 'No' in the 'Passed' category for both genders, educators can determine if specific demographics require additional academic support.



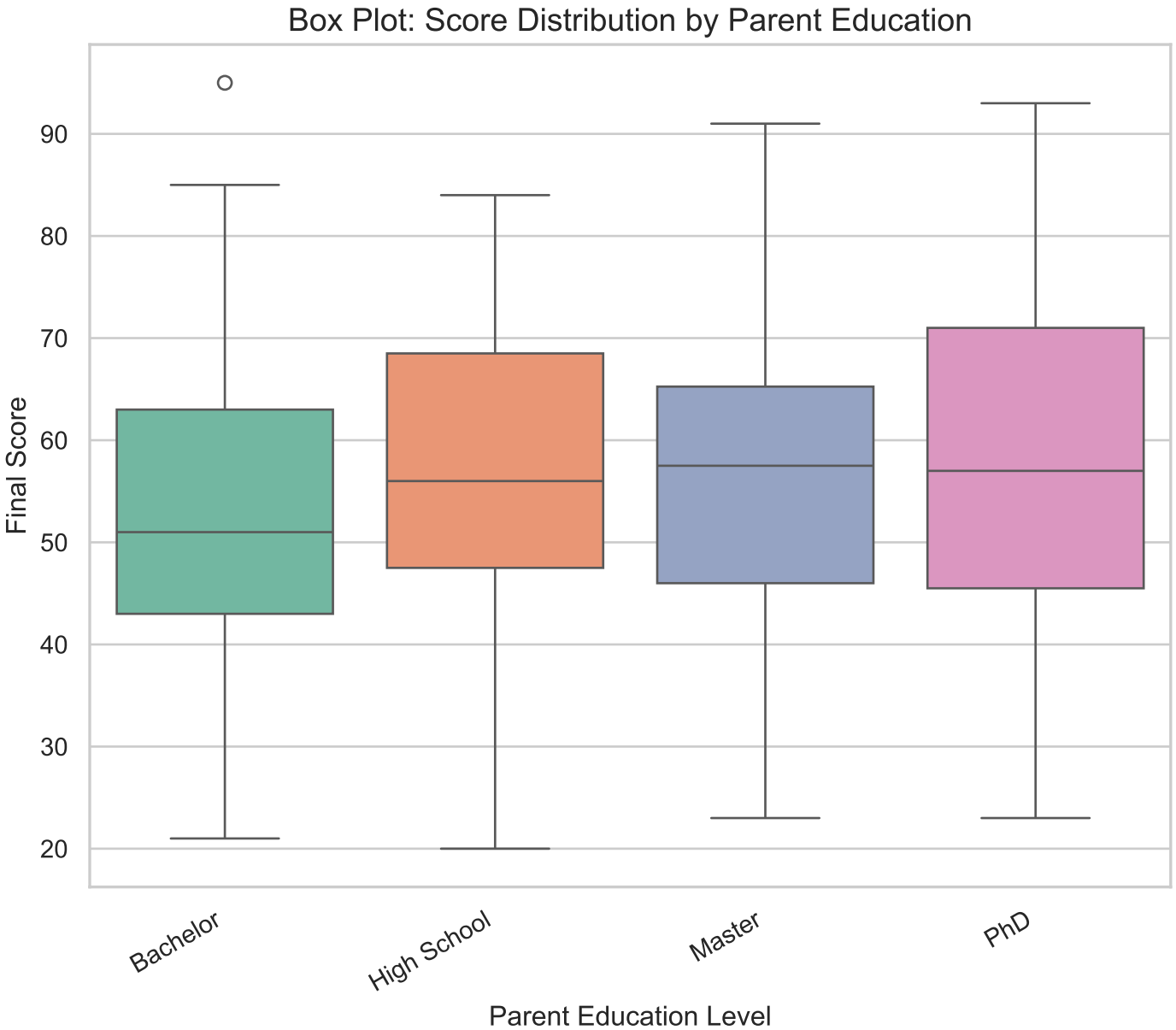
3. Correlation: Study Effort vs. Results

The scatter plot examines the relationship between weekly study hours and final exam scores. A positive trend (upward slope) suggests that increased study time directly impacts academic success. Points are color-coded by passing status, allowing for a clear visualization of the effort-reward threshold for passing the course.



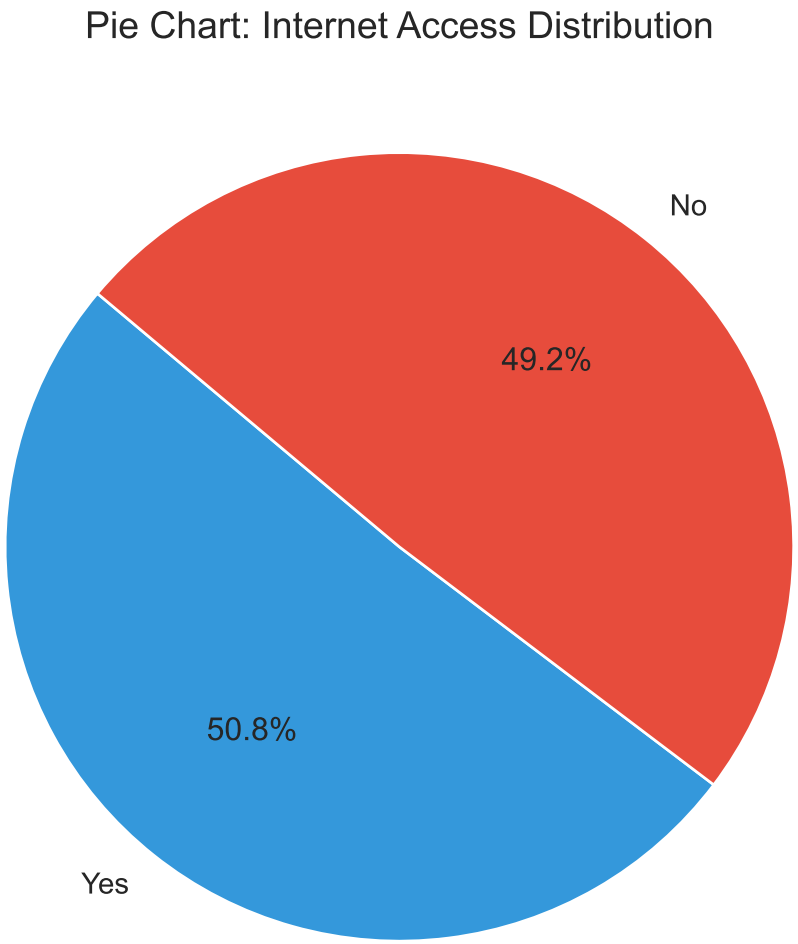
4. Impact of Socio-Economic Factors

This box plot displays the distribution of final scores across different levels of parental education. It provides insights into how a student's home environment might correlate with academic outcomes. The box represents the interquartile range (IQR), while the horizontal line within it marks the median score for each group.



5. Resource Accessibility: Internet Access

The pie chart represents the proportion of students who have access to the internet. In modern education, internet access is a critical variable as it dictates a student's ability to access online resources, research materials, and educational platforms. This breakdown highlights the digital divide within the student population.



6. Comprehensive Statistical Summary

```
NUMERIC PERFORMANCE METRICS:

AGE:
  - Mean: 16.98
  - Median: 17.00
  - Std Dev: 1.43

STUDY_HOURS_PER_WEEK:
  - Mean: 15.31
  - Median: 15.00
  - Std Dev: 8.57

ATTENDANCE_RATE:
  - Mean: 76.38
  - Median: 76.50
  - Std Dev: 13.82

PREVIOUS_SCORE:
  - Mean: 62.99
  - Median: 64.00
  - Std Dev: 18.94

FINAL_SCORE:
  - Mean: 55.98
  - Median: 56.00
  - Std Dev: 15.37

DEMOGRAPHIC MODES (MOST COMMON):
  - Gender: Female (Freq: 273)
  - Parent_Education: Master (Freq: 100)
  - Internet_Access: Yes (Freq: 254)
  - Passed: Yes (Freq: 323)
```

This summary table provides a quantitative overview of the dataset. Central tendency measures (Mean and Median) offer insights into average performance, while Standard Deviation indicates the level of variability among students. Demographic modes highlight the most common characteristics of the student body.